Paleomagnetism and U–Pb geochronology of the Lac de Gras diabase dyke swarm, Slave Province, Canada: implications for relative drift of Slave and Superior provinces in the Paleoproterozoic[1]

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Abstract: Lac de Gras diabase dykes trend north to NNE across the central Slave Province of the Canadian Shield. U–Pb baddeleyite ages of 2023 ± 2 and 2027 ± 4 Ma are interpreted as dyke emplacement ages. These ages are similar to that of the Booth River igneous complex, exposed along the margins of Kilohigok Basin near the northern end of the dyke swarm. Ten paleomagnetic sites (from four to six dykes) yield a mean paleopole at 11.8°N, 92.1°W (dm = 8.4°, dp = 6.0°). A positive baked contact test where a Lac de Gras dyke crosscuts a NE-trending Malley dyke demonstrates that this pole is primary. It represents the first key Paleoproterozoic pole from the Slave Province and, hence, the first Paleoproterozoic Slave pole suitable for reconstructing paleocontinents. Although a direct comparison is not available with precisely dated paleopoles of identical age from other Archean cratons, a comparison is made with a sequence of precisely dated poles from Superior Province dyke swarms, including those 40–50 million years older and 25 million years younger. It yields two options depending on the relative magnetic polarity assumed for data from the two cratons. The two cratons were either at similar latitudes, but not in their present relative orientations, when the swarms were emplaced, or separated in latitude by ~40°–60°. In either case, they may have drifted separately or formed part of a single (super)continent that subsequently broke up with the two cratons drifting separately to attain their present configuration. Additional key paleopoles are required to distinguish between these interpretations.

Résumé : Les dykes de diabase du Lac de Gras ont une direction N à NNE à travers le centre de la province des Esclaves du Bouclier canadien. Les âges U–Pb, sur de la baddeleyite, de 2023 ± 2 et de 2027 ± 4 Ma, sont interprétés comme l'âge de la mise en place des dykes. Ces âges sont semblables à ceux du complexe igné de Booth River, lequel affleure le long des bordures du bassin Kilohigok à proximité de l'extrémité nord de l'essaim de dykes. Dix sites paléomagnétiques (de quatre à six dykes) ont donné un paléopôle moyen à 11,8° N, 92,1° W (dm = 8,4°, dp = 6,0°). Un test positif du contact, là où le dyke de Lac de Gras recoupe un dyke Malley, de direction NE, démontre que ce pôle est primaire. Il représente le premier pôle clé paléoprotérozoïque de la province des Esclaves et donc le premier pôle paléoprotérozoïque de la province des Esclaves qui convient pour la reconstruction des paléocontinents. Bien qu'il ne soit pas possible de le comparer directement avec des paléopôles d'âge identique datés avec précision d'autres cratons archéens, il est comparé à une séquence de pôles datés avec précision d'un essaim de dykes de la province du Supérieur, incluant ceux 40–50 Ma plus vieux et ceux 25 Ma plus jeunes. Cette comparaison donne deux options selon la polarité magnétique relative assumée pour les données des deux cratons. Les deux cratons étaient soit à des latitudes semblables, mais n'avaient pas leur orientation relative actuelle, lorsque les essaims ont été mis en place, soit séparés par une différence de latitude d'environ 40°–60°. Dans les deux cas, ils pourraient avoir dérivé séparément ou formé une partie d'un (super)continent qui se serait disloqué par la suite; les deux cratons auraient alors dérivé séparément pour atteindre leur configuration actuelle. D'autres paléopôles clés sont requis pour trancher entre ces interprétations.

[Traduit par la Rédaction]

Introduction

Laurentia (Fig. 1; Hoffman 1988) is composed of several Archean cratonic blocks or provinces separated by late Paleoproterozoic orogenic belts (e.g., Trans-Hudson Orogen) that are interpreted by most geologists and geophysicists to mark collisional zones between the Archean blocks (e.g., Gibb et al. 1983; Hoffman 1988; Ross et al. 1991; Bleeker 2003). This interpretation requires relative movement of the Archean cratons prior to their amalgamation into Laurentia by

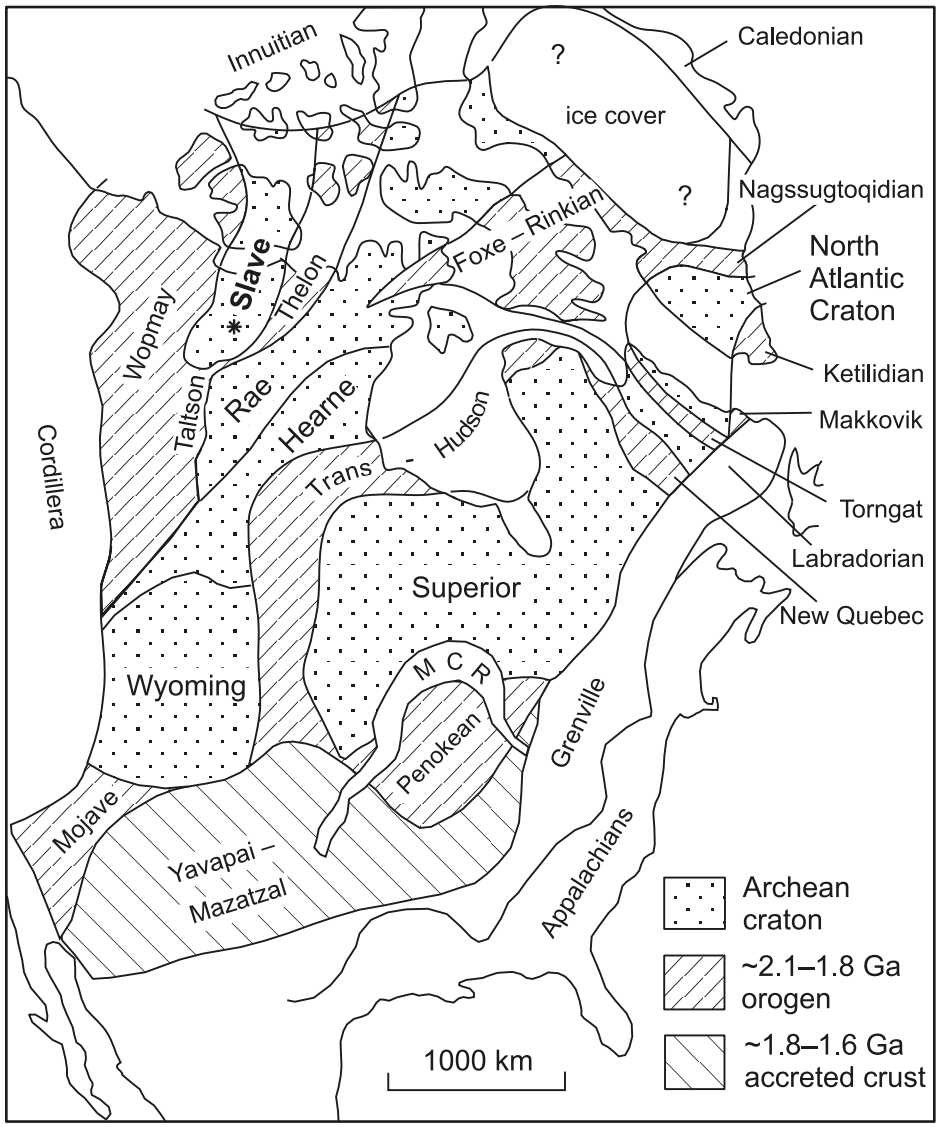
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Fig. 1. Tectonic map of Laurentia (modified after Hoffman 1988) with Greenland (upper right) shown in a pre-Mesozoic reconstruction. The star locates the Lac de Gras sampling area. MCR, Mid-Continent Rift.



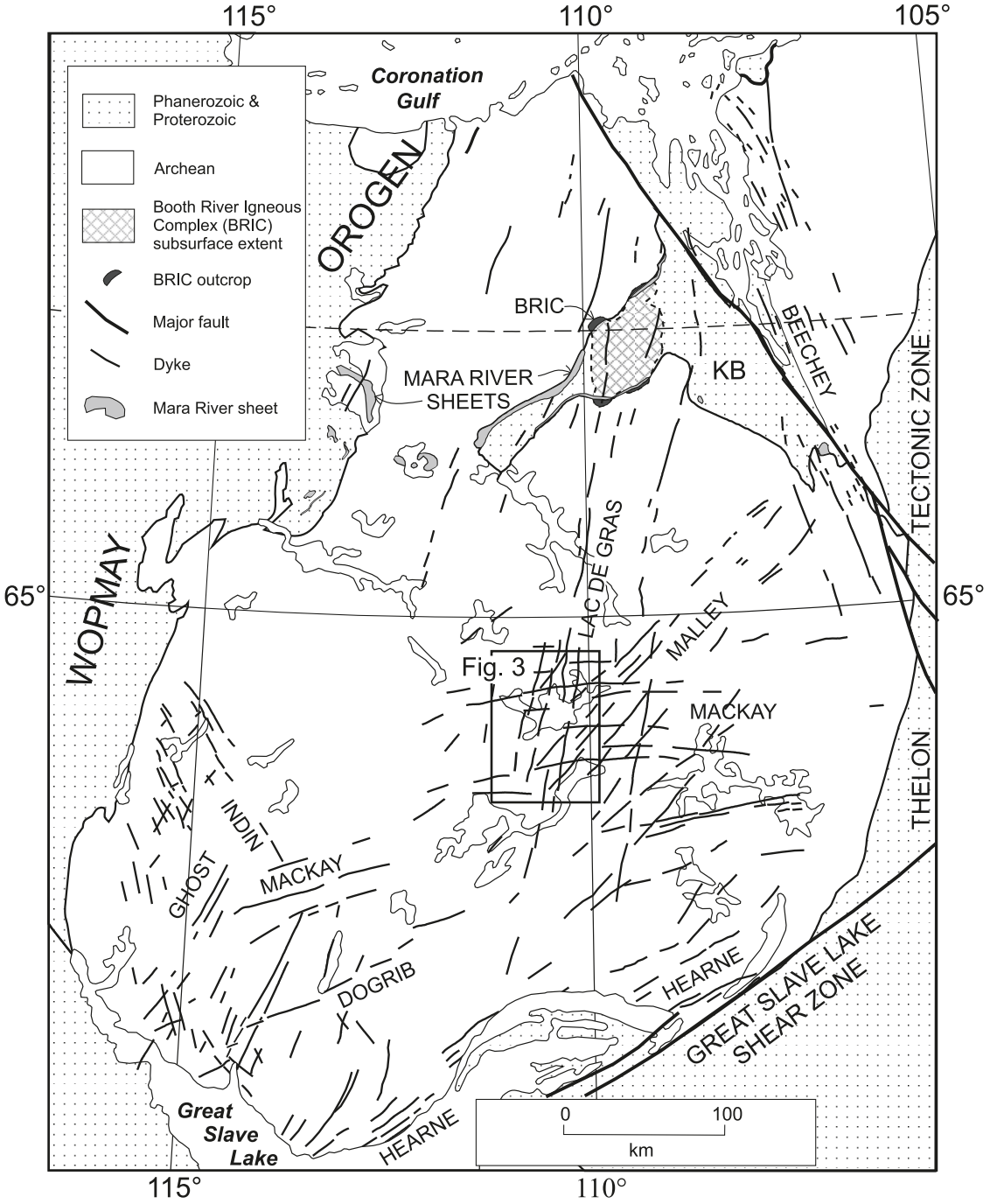
the late Paleoproterozoic. However, to date, conclusive paleomagnetic evidence for relative movement in the Paleoproterozoic among the Archean blocks that form Laurentia has been difficult to establish.

Several early attempts were made using apparent polar wander paths (APWPs) to establish whether the Slave and Superior provinces of the Canadian Shield drifted relative to one another during the Paleoproterozoic. Pullaiah and Irving (1975), McGlynn et al. (1975), Irving and McGlynn (1976), and Roy and Lapointe (1976) proposed that Paleoproterozoic paleopoles for the Superior and Slave provinces define a single APWP, indicating that the two provinces drifted together in their present configuration throughout the time interval. Piper (1987) concluded, mainly from paleomagnetic data, that all the shield areas in the world formed part of a "Proterozoic supercontinent" (with Superior and Slave provinces in their present relative configuration) from the late Archean to the Mesoproterozoic. On the other hand, Burke et al. (1976), Cavanaugh and Seyfert (1977), Irving et al. (1984), and Dunsmore and Symons (1990) proposed divergent APWPs for the Superior and Slave provinces prior to ca. 1750–1650 Ma and concluded that the two blocks had drifted separately.

The debate concerning Slave and Superior APWPs subsided after Roy (1983) and Buchan et al. (1993) pointed out that dating of Precambrian paleopoles was too uncertain to permit proper sequencing of poles along APWPs. Ages assigned to rock units, such as mafic intrusions and sedimentary sequences that are most often used for Precambrian paleomagnetic studies, typically had uncertainties of hundreds of millions of years. Dating and paleomagnetic studies were seldom carried out at the same sampling locality. Rigorous tests to establish whether remanences are primary or secondary were rarely conducted. In addition, relative magnetic polarities were often uncertain, and some early studies failed to completely isolate superimposed magnetic components because of now-outdated demagnetization procedures.

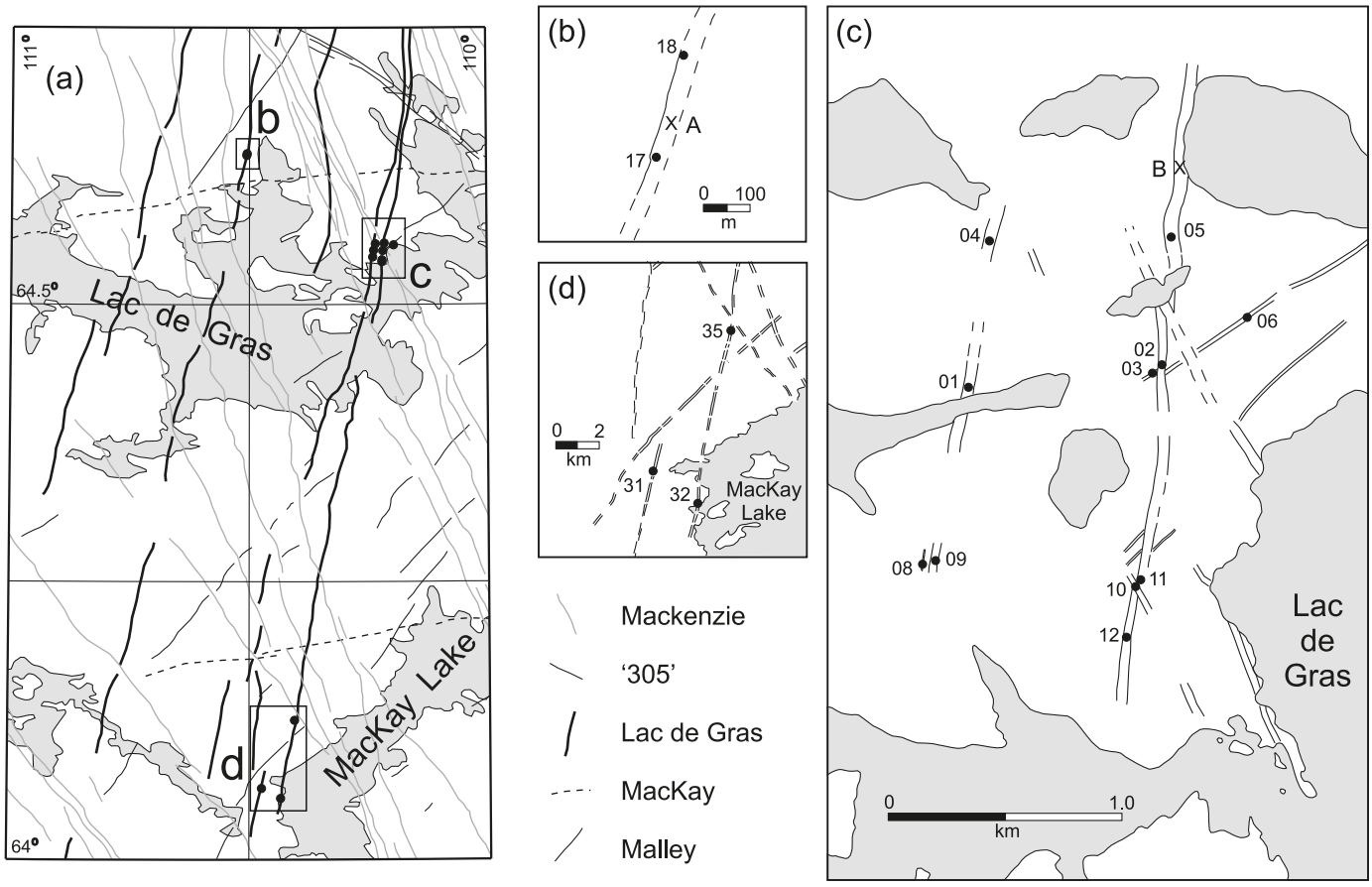
With the advent of high-precision U–Pb dating of badde-

Fig. 2. Paleoproterozoic dyke swarms of the Slave Province (modified after Buchan and Ernst 2004), from youngest to oldest, include ca. 1884 Ma NE-trending Ghost dykes (Atkinson 2004; Davis and Bleeker 2007; Bleeker et al. 2008*a*), ca. 1902 Ma ENE-trending Hearne dykes (Pietrzak 2003; Bleeker et al. 2008*b*), ca. 2027–2023 Ma north-trending Lac de Gras dykes (this study), ca. 2108 Ma NW-trending Indin dykes (McGlynn and Irving 1975; Davis and Bleeker 2007), ca. 2190 Ma ENE-trending Dogrib dykes (McGlynn and Irving 1975; LeCheminant et al. 1997), ca. 2210 Ma east-trending MacKay (or X) dykes (McGlynn and Irving 1975; Fahrig et al. 1984; LeCheminant et al. 1995), and ca. 2230 Ma NE-trending Malley dykes (Frith 1987; LeCheminant et al. 1995). BRIC, Booth River igneous complex; KB, Kilohigok Basin.



leyite, it became possible to accurately date Precambrian mafic intrusions (Krogh et al. 1987; Heaman and LeCheminant 1993). Buchan and Halls (1990) and Buchan et al. (1994) described the use of combined paleomagnetic and U–Pb geochronological studies of diabase dyke swarms and the rigorous application of paleomagnetic field tests to determine whether remanences are primary. Buchan et al. (1994) introduced the term "key paleopole" for poles that are both precisely dated and well defined, and hence useful for reconstructing paleocontinents. Criteria for key poles are described in detail by Buchan (2007*b*), and briefly summarized as follows.

Fig. 3. Diabase dykes (after LeCheminant 1994) and sampling sites of the Lac de Gras – MacKay Lake area. Sampling areas are enlarged in (*b–d*). Letters A and B identify geochronological sampling sites; numbers identify paleomagnetic sites.



A key paleopole must satisfy two basic criteria based on age and quality. Firstly, the paleopole must be demonstrated to be primary and the rock unit precisely and accurately dated. For purposes of this study we require uncertainties ≤±10 million years. Secondly, the primary magnetic component must be properly isolated in laboratory demagnetization experiments, secular variation largely averaged out, and corrections made for any significant crustal tilting.

Only U–Pb and, in some instances, Ar–Ar ages have the precision and accuracy that are necessary for a key paleopole. The most widely used field test to establish the primary nature of a remanence in intrusive rocks is the baked contact test (Everitt and Clegg 1962). However, it is often applied incorrectly (see discussion in Buchan 2007*a*). A rigorous baked contact test must include stable remanences in the intrusion, adjacent baked host rocks, and more distant unbaked host rocks. In conducting studies of diabase dyke swarms, it is important that paleomagnetic and geochronological samples are obtained from the same dyke, as there are many instances where parallel or subparallel dyke swarms of quite different ages occur in the same geographical area (e.g., Buchan et al. 1993, 2007*a*).

In recent years, a substantial number of key Paleoproterozoic paleopoles have been obtained for the Superior Province (e.g., Bates and Halls 1990; Buchan et al. 1993, 1998, 2007*a*; Halls and Heaman 2000). They demonstrate that Superior Province APWPs used during the Slave–Superior debate in the 1970s and 1980s are grossly in error and, therefore, that the early comparisons should be discarded.

No key Paleoproterozoic paleopoles have previously been reported for the Slave Province. We report the first such result, based on precise U–Pb ages for the emplacement of the Lac de Gras dyke swarm and a paleomagnetic remanence for the swarm that has been determined to be primary using a positive baked contact test. The Lac de Gras paleopole is compared with a sequence of key Paleoproterozoic poles from the Superior Province in the 2.2 to 2.0 Ga time interval.

Geological setting

The Slave Province is one of several Archean cratons that amalgamated to form Laurentia (Fig. 1) by the late Paleoproterozoic (Hoffman 1988). The geographic relationship in earlier times among these cratons and with other cratons that now reside in other continents is largely unknown. However, it is the subject of much interest because of its bearing on important issues, such as the nature of early supercontinents, the extent of mineral-rich belts within such supercontinents, the supercontinent cycle, and the timing of the onset of plate tectonics (e.g., Gurnis 1988; Rogers 1996; Bleeker 2003; Condie and Pease 2008).

Numerous Paleoproterozoic diabase dyke swarms were emplaced into the Slave craton prior to the formation of

Laurentia (e.g., Buchan and Ernst 2004; Stubley 2005; Bleeker et al. 2007). In addition to the Lac de Gras dykes of the present study, several other Paleoproterozoic swarms intruded the southern and central Slave Province (Fig. 2), including the ca. 2230 Ma NE-trending Malley swarm, ca. 2210 Ma east-trending MacKay (or "X") swarm, ca. 2190 Ma ENE-trending Dogrib swarm, ca. 2108 Ma NW-trending Indin swarm, ca. 1902 Ma ENE-trending Hearne swarm, and ca. 1884 Ma NE-trending Ghost swarm. These swarms are all crosscut by the extensive Mesoproterozoic Mackenzie swarm (1267 ± 2 Ma; LeCheminant and Heaman 1989) that forms a giant fan over much of the northern Canadian Shield (Fahrig 1987; Ernst and Baragar 1992). Northwest-trending "305" dykes, which may be roughly coeval with the Mackenzie swarm on the basis of paleomagnetic evidence (K. Buchan in LeCheminant 1994), also crosscut several of the Paleoproterozoic swarms in the sampling area. Many of the Paleoproterozoic swarms may represent extensional events associated with progressive breakup of a large ancestral craton (LeCheminant et al. 1996; Davis and Bleeker 2007). It is likely that remnants of these swarms exist on other, as yet unidentified, cratonic blocks that were once contiguous with the Slave craton.

During the ca. 1.9–1.8 Ga formation of Laurentia, the Slave craton was affected by collisional events along both its eastern and western margins. As a result, the Paleoproterozoic dyke swarms are more metamorphosed towards the Thelon Tectonic Zone on the east and Wopmay Orogen on the west (e.g., Henderson et al. 1990; Frith 1993). In the eastern Slave Province, the metamorphic grade of Malley and MacKay diabase dykes has been shown to increase systematically towards the Thelon Tectonic Zone (Henderson et al. 1990, fig. 3). Rb–Sr biotite ages in this region indicate that cooling through 300 ± 50 °C occurred at ca. 1735 Ma (Henderson et al. 1990, fig. 5). In the southwestern Slave Province, close to the Wopmay Orogen, it has been suggested (see Frith 1993, p. 32) that "Indin" dykes of different trends carry similar paleomagnetic directions as a result of metamorphic overprinting rather than as a result of them being of similar age, as had been proposed by McGlynn and Irving (1975). New U–Pb baddeleyite dating confirms a significant age difference between 1884 Ma "NE Indin" (now Ghost) and 2108 Ma "NW Indin" (now Indin) dykes (Bleeker et al. 2008*a*).

The Lac de Gras dyke swarm (Fig. 2) is distinctive in trend and geochemistry. It consists of an ~100 km wide array of ~010°-trending, steep to vertical diabase dykes (LeCheminant 1994). It extends north more than 300 km from Lac de Gras to the Booth River igneous complex (Roscoe et al. 1987; Hulbert 2005), a large layered mafic–ultramafic–felsic intrusion overlain by Goulburn Group sedimentary rocks of the Kilohigok Basin. Dykes of similar trend to the north of the Booth River intrusion (Fahrig 1987; Buchan and Ernst 2004) may also be part of the Lac de Gras swarm (Fig. 2). South of Lac de Gras, the swarm can be traced to the vicinity of Snap Lake at latitude 63.5°N (Stubley 2005).

Geochemical characteristics of Lac de Gras dykes are distinct from those of other Proterozoic diabase dykes in the southern and central Slave Province (Wilkinson et al. 2001; Ernst and Hulbert 2003). Lac de Gras dykes plot in the alkaline field of a total alkali–silica diagram, whereas other Proterozoic swarms near Lac de Gras are subalkaline in composition (Wilkinson et al. 2001). Lac de Gras dykes have higher light rare-earth element content and steeper rare-earth element patterns than those of other swarms. Their chemistry is consistent with alkalic intraplate basaltic magmatism, whereas older Paleoproterozoic Malley and MacKay dykes and younger Mesoproterozoic Mackenzie and "305" dykes have signatures typical of continental tholeiitic magmatism (Wilkinson et al. 2001).

Two Lac de Gras dykes have been dated in the present study at 2023 ± 2 and 2027 ± 4 Ma (U–Pb baddeleyite), as discussed later in the text. These ages are similar to U–Pb zircon ages of 2023 $^{+4}_{-2}$Ma for the felsic phase (Roscoe et al. 1987) and 2026 ± 1 and 2025 ± 1 Ma for the mafic phase (Davis et al. 2004) of the Booth River igneous complex. Therefore, the Lac de Gras dyke swarm and Booth River intrusion are coeval and part of a magmatic event that occurred shortly before collision(s) along the eastern margin of the Slave Province that resulted in formation of the 2.0–1.9 Ga Thelon Tectonic Zone (Henderson et al. 1990). An ENE-trending dyke in the Great Slave Lake Shear Zone (Fig. 2) along the southern margin of the Slave Province has been dated at 2038 ± 3 Ma (Pehrsson et al. 1993), only a few million years older than the Lac de Gras dykes and Booth River intrusion.

Sampling and laboratory procedure

Geochronological and paleomagnetic sampling of Lac de Gras dykes was carried out at sites in the Lac de Gras – MacKay Lake area (Fig. 3). Here, Lac de Gras dykes intersect older Malley and MacKay dykes and are crosscut by younger Mackenzie dykes.

U–Pb ID–TIMS geochronology

The two U–Pb geochronology sites (A and B in Fig. 3) are in dykes that were also sampled for paleomagnetism. At the western site (A), a medium-grained gabbro suitable for baddeleyite mineral separation was obtained 15 m from the margin of a 40 m wide dyke. At the eastern site (B), a similar sample was collected 23 m from the margin of an ~50 m wide dyke.

Separation of baddeleyite followed conventional techniques of crushing, Wilfley table concentration, heavy liquids, and magnetic separation. Isotope dilution – thermal ionization mass spectrometry (ID–TIMS) procedures for U and Pb isotopic analysis are described in Parrish et al. (1987). Crystals selected for analysis were grouped into fractions based on quality, determined primarily by uniform colour and absence of turbidity, fractures, inclusions, and attached grains (Table 1). Treatment of analytical errors is based on Roddick (1987).

Paleomagnetism

Paleomagnetic block samples (usually five per site) were oriented using both magnetic and solar compasses at 13 Lac de Gras dyke sampling sites (Fig. 3). Because it is difficult to trace individual dykes across the sampling area, the 13 sites may be associated with four to six dykes (Fig. 3). Twelve of the Lac de Gras dyke sites are unbaked, whereas one (site 11) is baked by a Mackenzie dyke. Paleomagnetic

Table 1. U–Pb TIMS analyses on baddeleyite from the Lac de Gras dyke swarm.

Fraction[a]	Weight (μg)[b]	U (ppm)	Pb (ppm)[c]	$^{206}Pb/^{204}Pb$[d]	Pb (pg)[e]	$^{208}Pb/^{206}Pb$[f]	$^{207}Pb/^{235}U$[f]	$^{206}Pb/^{238}U$[f]	$^{207}Pb/^{206}Pb$[f]	$^{206}Pb/^{238}U$[g]	$^{207}Pb/^{235}U$[g]	$^{207}Pb/^{206}Pb$[g]	Disc[h]
Site A, western dyke (92LAAT38-1, Z2796; 64.633°N, 110.500°E)													
A-1, 100×40	8	248	88	2237	20	0.01	6.264±0.12	0.3646±0.11	0.12460±0.06	2003.9±3.7	2013.4±2.2	2023±2	1.11
A-2, 90×30	2	197	70	599	15	0.01	6.296±0.20	0.3649±0.15	0.12515±0.15	2005.3±5.0	2017.9±3.5	2031±5	1.47
A-3, 100×40	4	315	113	2024	15	0.01	6.298±0.11	0.3666±0.08	0.12460±0.05	2013.2±2.9	2018.1±1.9	2023±2	0.57
Site B, eastern dyke (92LAAT37-1, Z2898; 64.617°N, 110.483°E)													
B-1, 120×50	9	299	107	2727	23	0.02	6.279±0.12	0.3643±0.10	0.12500±0.06	2002.8±3.6	2015.6±2.1	2029±2	1.49
B-2, 120×80	19	249	87	3403	30	0.02	6.129±0.12	0.3548±0.10	0.12528±0.05	1957.7±3.5	1994.4±2.1	2033±2	4.28
B-3, 110×50	18	261	92	9162	11	0.01	6.200±0.10	0.3602±0.09	0.12486±0.03	1982.9±3.1	2004.5±1.8	2027±1	2.51

[a]Non-magnetic at a side slope given in degrees on a Frantz isodynamic magnetic separator operating at 1.8 A; sizes of fragments in μm, i.e., 100 × 50; average length and breadth, thickness is generally 1/3 of the breadth.
[b]Error on weight = ±1 μg.
[c]Radiogenic Pb.
[d]Measured ratio corrected for spike and Pb fractionation of 0.09 ± 0.03% per AMU (atomic mass units).
[e]Total common Pb in analysis corrected for fractionation and spike.
[f]Corrected for blank Pb and U, common Pb, errors quoted are 1σ in percent.
[g]Age errors quoted are 2 SE in Ma.
[h]Discordance in percent along a discordia to origin.

sites 17 and 18 are located on one of the dated dykes, a short distance south and north, respectively, of geochronology sampling site A (Fig. 3*b*). Paleomagnetic site 05 is ~300 m south of geochronology site B on the other dated dyke (Fig. 3*c*).

Baked contact tests were carried out at two localities (see Fig. 3*c*). In one test, a Lac de Gras dyke (site 11) is baked by a crosscutting NNW-trending Mackenzie dyke (site 10). Site 12 represents the unbaked Lac de Gras dyke farther south. In the second test, a NE-trending Malley dyke (site 03) is baked by a Lac de Gras dyke (site 02). The unbaked Malley dyke was sampled at site 06.

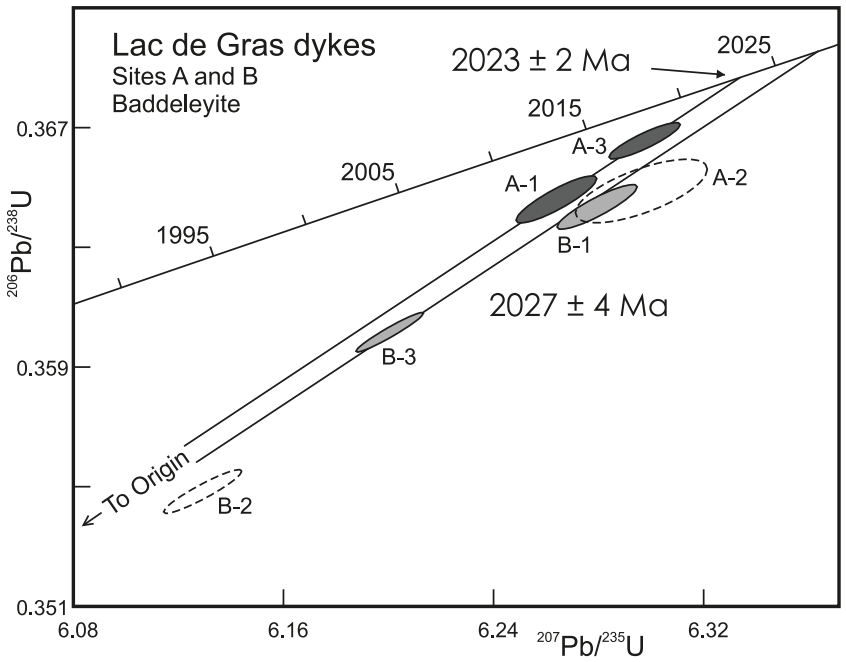
Cylindrical specimens 2.2 cm long and 2.5 cm in diameter were drilled from each block sample. Specimens were measured using a Schonstedt DSM-1 spinner magnetometer, and were subjected to stepwise alternating field (AF) demagnetization, stepwise thermal demagnetization, or a combination of AF followed by thermal cleaning. The AF and thermal experiments were carried out with a Schonstedt GSD-1 demagnetizer and a modified Schonstedt furnace, respectively. All experimental work was conducted in a shielded room with residual magnetic field <3000 nT.

Petrography

Lac de Gras dykes are fresh and undeformed, although some dyke rocks display weak deuteric alteration. Chilled margins contain abundant plagioclase and clinopyroxene microphenocrysts and grade into fine- to medium-grained gabbros with ophitic to subophitic textures. The gabbros have an unusually high modal content of Fe–Ti oxides that imparts a distinctive patchy look to the rocks. Petrographically, the gabbros are essentially composed of plagioclase, clinopyroxene, altered olivine, amphibole, and Fe–Ti oxides. Zoned clinopyroxene grains poikilitically enclose plagioclase laths and altered olivine. Centres of zoned plagioclase laths are variably saussuritized and rims have a sharp break to sodic compositions adjacent to interstices. Primary amphibole partly rims clinopyroxene and altered olivine, and is a minor interstitial mineral. Fe–Ti oxide content ranges up to 6 modal percent, primarily as blocky, coarse-lamellar titanomagnetite–ilmenite grains, some with feathery ilmenite-rich rims. Seams of fine skeletal magnetite form irregular webs throughout altered olivine. Elongate ilmenite blades are associated with late crystallizing minerals in interstices.

Interstices contain a diverse assemblage of late-stage primary minerals, particularly those rich in Fe, Ti, Zr, and radioactive elements. Quartz, alkali feldspar, amphibole, Fe-rich biotite, and chlorite fill straight-sided interstices, where they are associated with ilmenite and apatite needles, minor sulphides, and various radioactive trace minerals. Amphibole, biotite, and a few apatite grains are Cl-bearing. Sulphides include pyrite, pyrrhotite, chalcopyrite, sphalerite, galena, and molybdenite. Radioactive minerals are titanite, allanite, baddeleyite, rare zirconolite, and zircon. The complexly zoned allanite grains and irregularly altered zircon crystals are locally abundant. Euhedral baddeleyite grains in the two dated samples are not altered; many of the crystals are finely striated and multiple crystal groupings are common. Scanning electron microscope images show unusual sector zoning in some crystals. Late-stage deuteric altera-

Fig. 4. U–Pb concordia diagram for baddeleyite fractions from two Lac de Gras dykes. Data points in dashed lines are not used for age calculations.



tion, apparent in a few dykes, is associated with thin amphibole–chlorite–titanite veinlets and localized alteration of mafic minerals to secondary amphibole, chlorite, and carbonate. Baddeleyite grains subject to late alteration effects have developed only a thin fringe of secondary zircon.

U–Pb geochronological results

U–Pb data are presented in Table 1 and Fig. 4. The Lac de Gras dyke at site A yielded ~200 baddeleyite grains averaging 40 μm × 100 μm from processing 9 kg of sample. Uranium concentrations in the three analyzed fractions range from 197 to 315 ppm. Fraction A-1 (1.1% discordant) and A-3 (0.6% discordant) have identical $^{207}Pb/^{206}Pb$ ages that yield a weighted average of 2023 ± 2 Ma. The close agreement in $^{207}Pb/^{206}Pb$ ages suggests that discordance was the result of recent Pb loss. Thus, 2023 ± 2 Ma, based only on fractions A-1 and A-3, is considered to be the best estimate for emplacement of this dyke.

Baddeleyite fractions from the Lac de Gras dyke at site B have a small range of U concentrations (249–299 ppm). More than 500 good quality baddeleyite grains averaging ~100 μm in length were recovered from a 9 kg sample. The three analyses are not perfectly collinear but again suggest recent Pb loss (Fig. 4). The most concordant fractions (B-1 and B-3) have similar $^{207}Pb/^{206}Pb$ ages of 2027 ± 1 and 2029 ± 1 Ma. Fraction B-2, which is more discordant, has a $^{207}Pb/^{206}Pb$ age of 2033 ± 2 Ma. The weighted average age of 2027 ± 4 Ma for fractions B-1 and B-3 is interpreted as the best estimate of the emplacement age of this dyke.

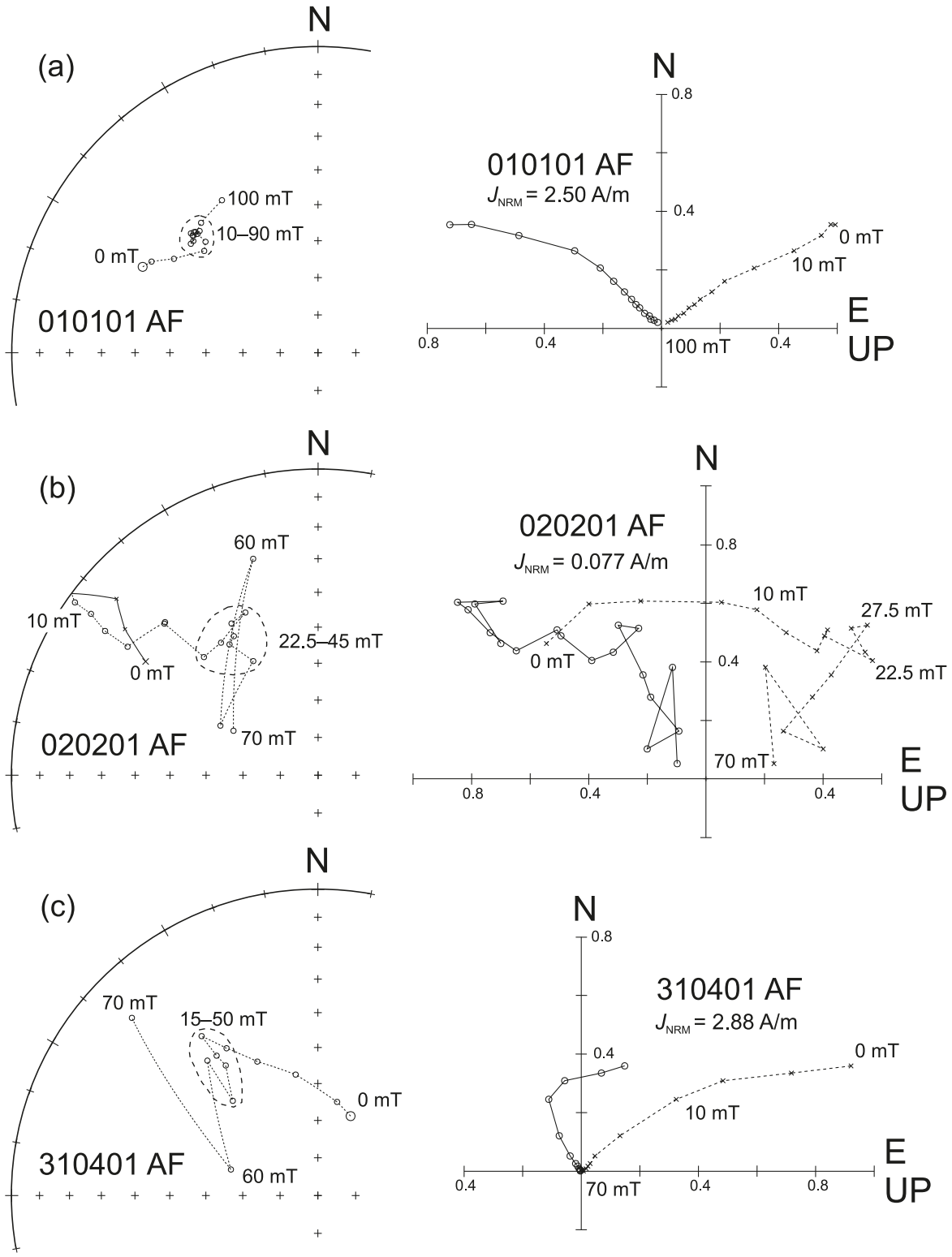
Paleomagnetic results

Samples from two of the 12 paleomagnetic sampling sites (18 and 35) collected in unbaked Lac de Gras dykes yielded highly scattered or unstable remanence directions. The remaining 10 Lac de Gras sites yielded internally consistent directions of remanent magnetization after AF, thermal, or combined AF–thermal demagnetization. Examples of demagnetization experiments are illustrated in Figs. 5–7. Paleomagnetic directions are summarized in Fig. 8 and Table 2.

The characteristic (or A) magnetization component in the Lac de Gras dykes is directed up to the NNW. For some samples, AF demagnetization was most efficient at eliminating other, generally less stable, components (Fig. 5*a*). In other cases, however, the A component was only poorly defined on AF cleaning (Figs. 5*b*, 5*c*). Thermal demagnetization or combined AF–thermal treatment often proved to be more efficient than AF cleaning at isolating the A magnetization. In some samples, thermal demagnetization yielded well-defined A endpoints (Fig. 6). In many others, AF demagnetization to between 5 and 15 mT followed by thermal cleaning gave excellent results (Fig. 7). Blocking temperatures of the A component are usually high, ~520–570 °C (Figs. 6, 7), suggesting that low-Ti titanomagnetite is the dominant carrier of this remanance.

The A remanence has a mean direction of $D = 338.2°$, $I = -55.3°$ ($\alpha_{95} = 5.9°$, $k = 68$) (Fig. 8; Table 2) based on data from 10 Lac de Gras sites (sites 01, 02, 04, 05, 08, 09, 12, 17, 31, and 32). As noted earlier, the 10 sites are located on four to six individual dykes. Therefore, much, but not necessarily all, of the secular variation is averaged out. A mean direction based on dyke data could not be calculated because it is not possible to trace individual dykes across the sampling area (Fig. 3*a*). Malley dyke site 03, which was baked by an adjacent Lac de Gras dyke (site 02), carries a stable A remanence ($D = 329°$, $I = -52°$) that contrasts with the characteristic remanence ($D = 127°$, $I = -60°$) observed at unbaked Malley dyke site 06 (Fig. 8). Lac de Gras site 11, which was baked by a crosscutting Mackenzie dyke (site 10), carries a Mackenzie overprint ($D = 239°$, $I = 28°$),

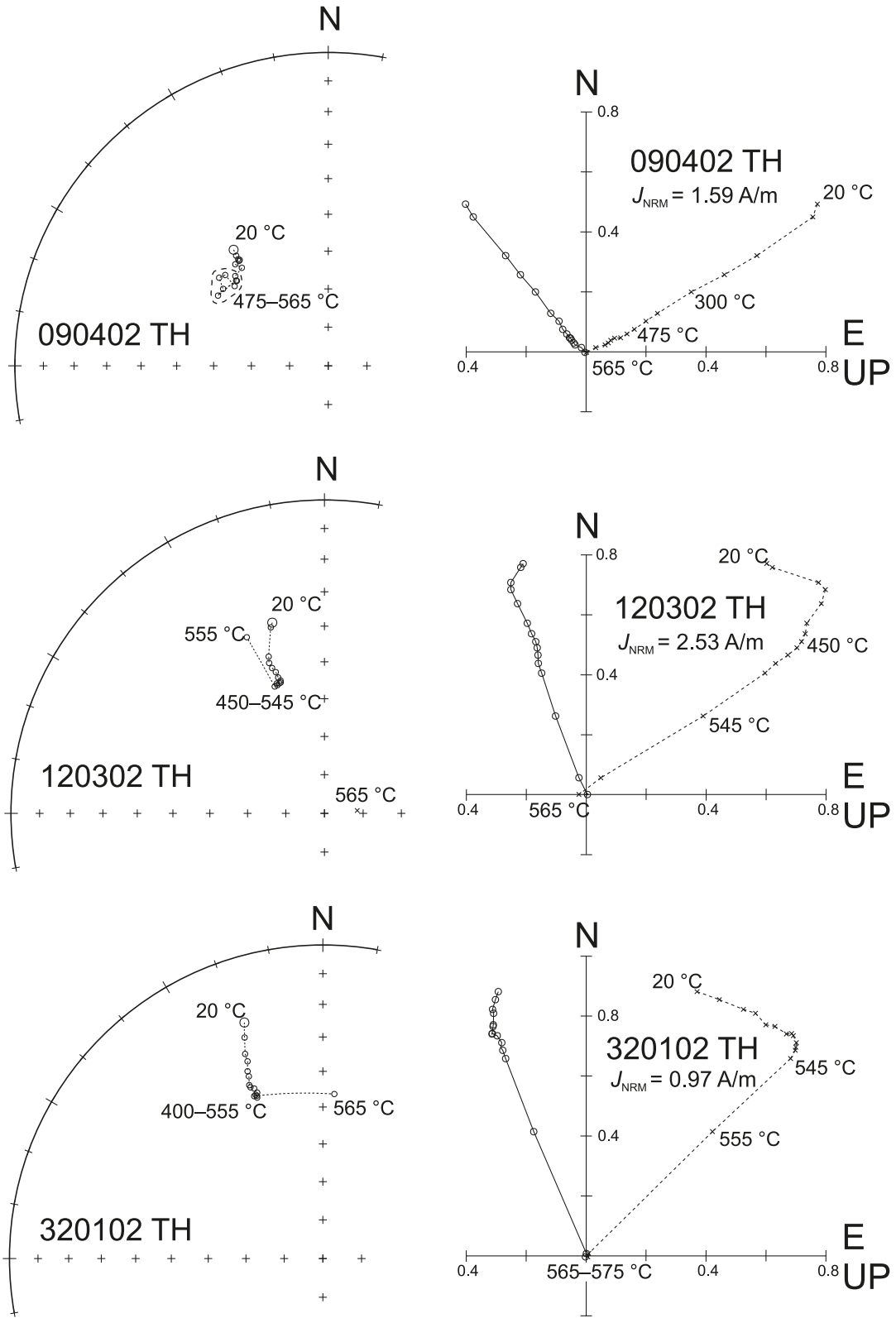
Fig. 5. Paleomagnetic results for Lac de Gras dykes based on AF demagnetization. Six-digit specimen numbers identify sampling site (first two digits), sample (third and fourth digits), and specimen (last two digits). Paleomagnetic directions are plotted on equal-area projections with closed (open) symbols indicating down (up) directions. Circles (crosses) on vector diagrams represent projection onto the horizontal (vertical) plane. J_{NRM} is the initial intensity of natural remanent magnetization.



as illustrated in Fig. 8. The baked contact tests are discussed in detail in the following subsection.

In addition to the A component, other, usually less stable, components are superimposed at most sampling sites (Figs. 5–7). In some cases, they are directed close to that of the present Earth's field. Usually they are quite scattered. A

Fig. 6. Paleomagnetic results for Lac de Gras dykes based on thermal demagnetization. Symbols as in Fig. 5.



down SE (or B) component is observed in several samples from Lac de Gras dyke sites 04, 09, 17, and 31. It has lower blocking temperatures than the A component, but its coercivities are lower than, higher than, or overlapping those of the A component. As explained in the "Discussion," we interpret this B component to be a secondary overprint acquired long after emplacement of the Lac de Gras dyke swarm. The B component is also present at Malley dyke site 06 (see baked contact test discussion in the following subsection). At the two sites (18 and 35) with scattered or

Fig. 7. Paleomagnetic results for Lac de Gras dykes based on combined AF–thermal demagnetization. Symbols as in Fig. 5.

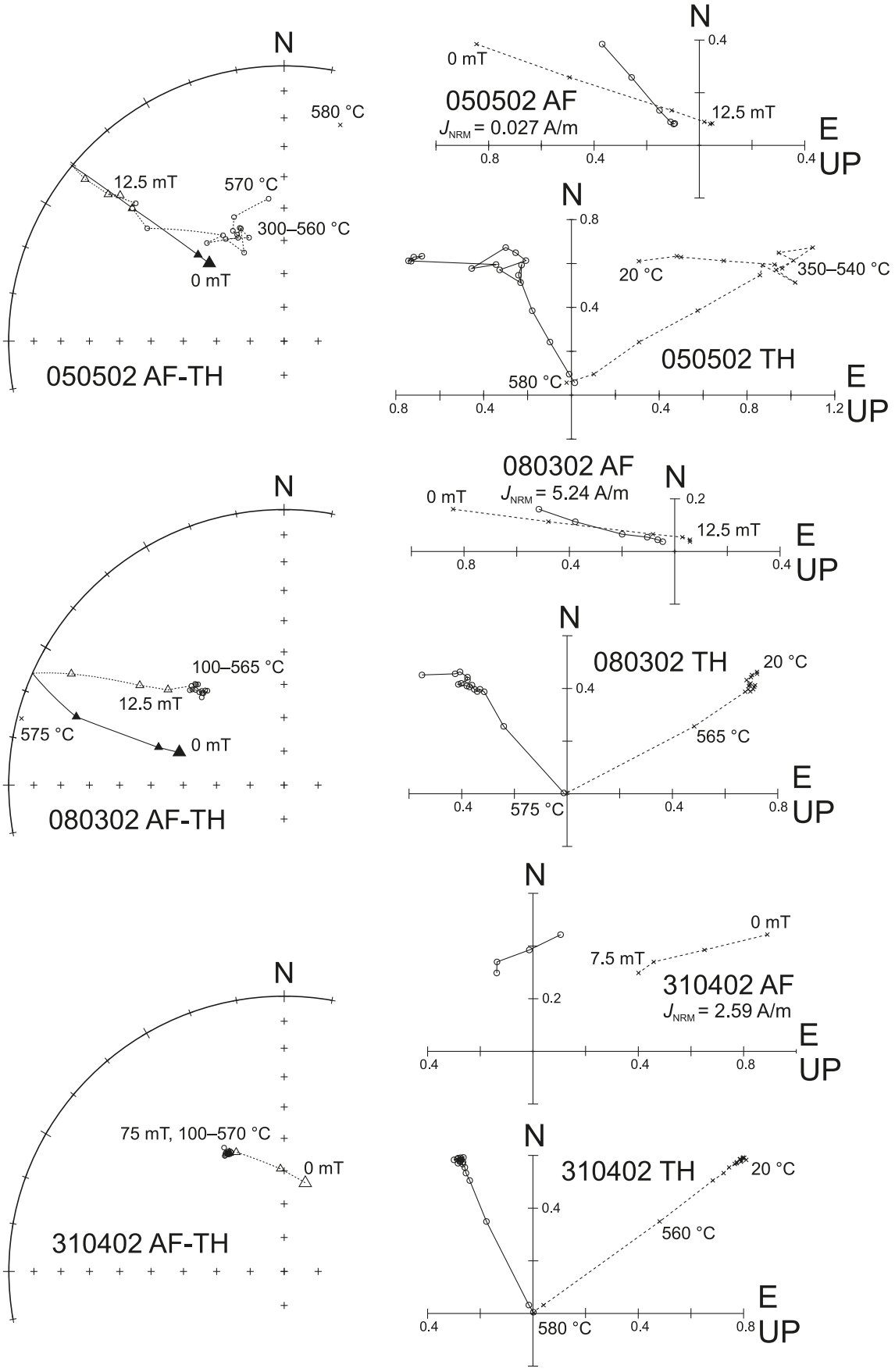
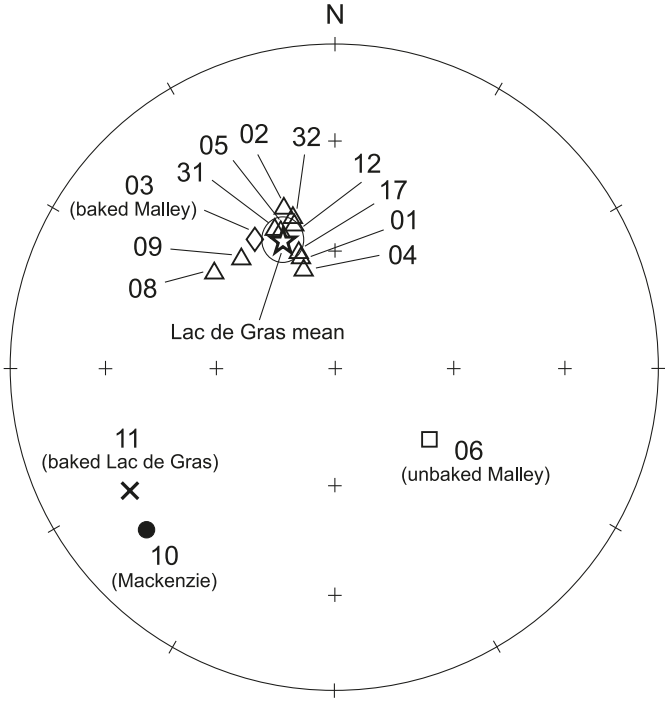


Fig. 8. Paleomagnetic site mean directions for Lac de Gras dykes and baked contact tests (see Figs. 9, 10). Data are plotted on equal-area projections with open symbols indicating up directions and the closed circle and X indicating down directions. Triangles are Lac de Gras A site means. The star is the mean of 10 Lac de Gras A site directions. The diamond (site 03) is the direction from a Malley dyke in the baked contact zone of a Lac de Gras dyke. The open square (site 06) is the direction from an unbaked site in the same Malley dyke. The closed circle (site 10) is the direction from a Mackenzie dyke, where it baked a Lac de Gras site (X, site 11).



unstable remanence directions, large low-temperature components are superimposed on smaller magnetite components in some samples. The maximum blocking temperatures of the low-temperature component fall in the 300–350 °C range suggesting that pyrrhotite is the remanence carrier.

Baked contact tests

A baked contact test (Fig. 9) was carried out where a >30 m wide 1267 Ma Mackenzie dyke (site 10) with a trend of 330° crosscuts a Lac de Gras dyke (sites 11 and 12) trending 010°. The Mackenzie dyke carries a remanence direction (D = 229°, I = 25°) typical of other dykes in this swarm (e.g., Robertson 1969). Two samples collected in the Lac de Gras dyke ~3.5 m from the Mackenzie dyke contact (site 11) also carry a Mackenzie direction (D = 239°, I = 28°). The A remanence direction (D = 355°, I = –52°) was obtained at site 12 in the same Lac de Gras dyke ~150 m from the Mackenzie dyke contact. This positive baked contact test demonstrates that the remanence of the Mackenzie dyke at site 10 is primary and that the A remanence of the unbaked Lac de Gras dyke (site 12) significantly predates emplacement of the Mackenzie dyke.

A second baked contact test (Fig. 10) was conducted where a Lac de Gras dyke (site 02) trending 015° crosscuts a Malley dyke (sites 03 and 06) with a trend of 045° (Fig. 10*a*). Although dyke contacts at the intersection are not exposed, a crosscutting relationship is inferred from the offset in the Malley dyke that occurred when the Lac de Gras dyke was emplaced (Fig. 10*a*). The Lac de Gras dyke at site 02 carries an A remanence direction (D = 343°, I = –47°), indistinguishable from the direction (D = 329°, I = –52°) of the baked Malley dyke (site 03) only a few metres away. The same Malley dyke sampled at site 06, ~500 m to the northeast of the Lac de Gras intersection, carries a stable, high blocking temperature component directed steeply up to the SE (D = 127°, I = –60°), and a lower temperature B component directed down to the SE (Fig. 10*e*). The B component has maximum blocking temperatures between 300 and 350 °C and relatively high coercivities, suggesting that it is carried by pyrrhotite. It is likely an overprint acquired after emplacement of both the Malley and Lac de Gras dykes (see "Discussion"). This positive baked contact test demonstrates that the A component of the Lac de Gras dyke is primary.

Discussion

Low coercivity or low blocking temperature magnetic components parallel to present Earth's magnetic field observed in some Lac de Gras samples are interpreted to be of recent origin. The origin of soft, scattered components is unclear, but it is likely that they are of secondary origin. The B component (directed down to the SE), which is superimposed with a higher blocking temperature component at Malley dyke site 06 and with the A component in several samples at Lac de Gras dyke sites 04, 09, 17, and 31, is also likely of secondary origin. Because it is roughly reversed to the Lac de Gras A component, it could be about the same age. However, a similar component, superimposed on more stable remanences, has also been observed in a number of previous paleomagnetic studies of Paleoproterozoic units from the southern Slave Province, where it is interpreted as an overprint component. For example, it has been reported from two east-trending MacKay dykes in the present sampling area (Fahrig et al. 1984), from the margin of ENE-trending Dogrib dykes in the Yellowknife area (McGlynn and Irving 1975), and from some Yellowknife sediments near diabase contacts (McGlynn and Irving 1975). The B component is also widely observed across Canada and Greenland as an overprint associated with cooling at the end of the "Hudsonian" Orogeny, when Archean cratons amalgamated to form Laurentia (see summary in Irving et al. 2004). These overprints are consistently of normal polarity and it is thought that they were acquired at ca. 1.7 Ga (Irving et al. 2004). They do not occur in younger units such as Mackenzie dykes.

The A magnetization of Lac de Gras dykes is established as primary based on the baked contact test at the Lac de Gras – Malley dyke intersection. The corresponding paleomagnetic pole falls at 11.8°N, 92.1°W (dm = 8.4°, dp = 6.0°) and is considered to be a key pole, the first key Paleoproterozoic pole from the Slave Province.

In the following discussion, we compare the 2027–2023 Ma Lac de Gras paleopoles from the Slave Province with data from the Superior Province, the only craton in the world for which a sequence of key Paleoproterozoic poles has been determined (e.g., Buchan et al. 2000, 2007*a*, 2007*b*). The Superior Province poles outline a rudimentary

Table 2. Paleomagnetic results.

Site	Dyke	Trend (°)	Width (m)	Latitude (°N)	Longitude (°W)	N (n_A, n_T)	D (°)	I (°)	k	α_{95} (°)
01	Lac de Gras	020	>40	64.5500	110.2325	5(5,0)	352.8	–60.9	63	17/28
02[a]	Lac de Gras	015	~30?	64.5505	110.2131	6(3,0)	342.6	–47.1	40	20
03[a]	Malley baked by Lac de Gras	045	11	64.5506	110.2137	4(4,0)	328.8	–51.8	120	8
04	Lac de Gras	005	>40	64.5555	110.2297	5(3,1)	342.6	–64.4	36	15
05	Lac de Gras	000	50	64.5564	110.2125	5(1,4)	338.7	–53.0	43	12
06[a]	Unbaked Malley	045	16	64.5532	110.2053	5(0,4)	126.9	–59.8	56	12
08	Lac de Gras	015	0.22	64.5429	110.2350	4(2,2)	308.7	–51.3	92	10
09	Lac de Gras	~015	>3	64.5429	110.2344	5(0,4)	319.8	–53.6	26	19
10[b]	Mackenzie	330	>30	64.5411	110.2161	2(2,0)	229.2	24.9		θ = 9
11[b]	Lac de Gras baked by Mackenzie	~015	~30	64.5410	110.2159	2(2,0)	239.4	28.3		θ = 4
12[b]	Lac de Gras	~015?	~30?	64.5389	110.2162	4(4,0)	354.7	–52.1	61	12
17	Lac de Gras	~015	~40	64.6358	110.5015	5(0,5)	343.6	–59.8	118	7
18	Lac de Gras	~015	~40	64.6375	110.5006	5(0,0)	—	—	—	—
31	Lac de Gras	010	16.7	64.0647	110.4695	5(0,3)	336.4	–51.7	286	7
32	Lac de Gras	0–015	17.5	64.0559	110.4311	5(0,5)	345.4	–51.3	119	7
35	Lac de Gras	015	36.7	64.1260	110.4003	5(0,0)	—	—	—	—

Note: Symbols: N, the number of oriented samples collected at the site; n_A and n_T, the number of samples used to calculate the site mean direction that were demagnetized by AF or thermal techniques, respectively; D and I, the mean declination and inclination, respectively, of the characteristic remanence; k, the precision parameter; α_{95}, the radius of the circle of 95% confidence, with the exception of sites 10 and 11 for which the angular separation (θ) between the two sample directions is given. At site 01, where sample directions were determined from remagnetization circle analysis (Bailey and Halls 1984), minimum and maximum values of α_{95} are given. The mean direction for the 10 stably magnetized Lac de Gras dyke sites is $D = 338.2°$, $I = –55.3°$ ($\alpha_{95} = 5.9°$, $k = 68$). The corresponding paleomagnetic pole is located at 11.8°N, 92.1°W (dm = 8.4°, dp = 6.0°), where dm and dp are the semi-axes of the 95% confidence ellipse.

[a]Sites 02, 03, and 06 constitute the baked contact test of Fig. 10.

[b]Sites 10, 11, and 12 constitute the baked contact test of Fig. 9.

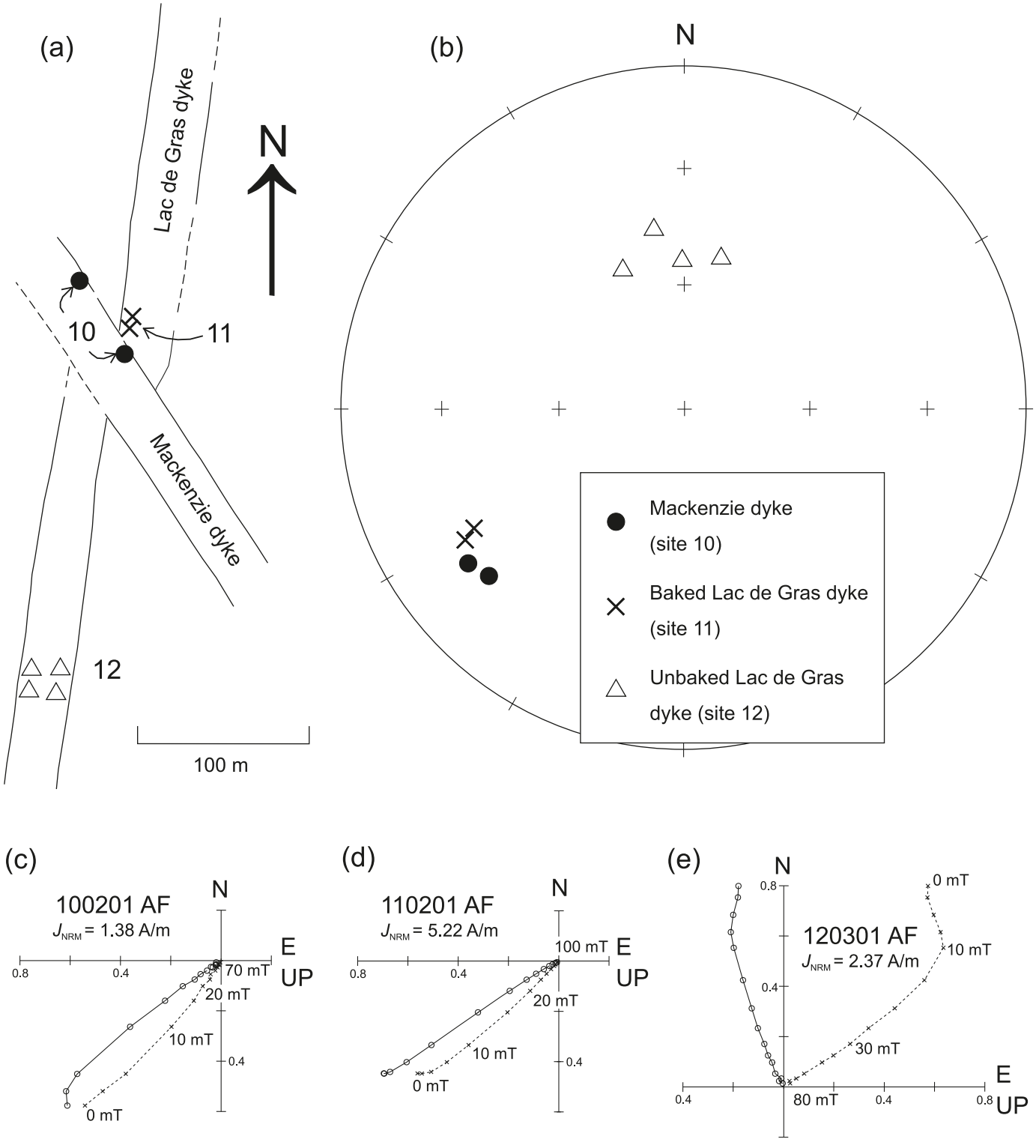
APWP that reflects drift of the craton and represents a reference track with which poles from other cratons can be compared (Buchan et al. 2007*b*). Although none of the Superior Province key poles is identical in age to the Lac de Gras pole, several fall in the 2.1–2.0 Ga time interval, sufficiently close in age to make a useful comparison.

Comparable data from other cratons in the 2.1–2.0 Ga interval are not yet available. Indeed, only two paleopoles from cratons other than the Slave and Superior provinces have field tests to establish that remanences are primary (Evans and Pisarevsky 2008). In both cases, however, the paleomagnetic data are complex and open to different interpretations. One pole is from Waterberg Group Sequence I sediments and 2054 ± 4 Ma (Dorland et al. 2006) lavas in the Kaapvaal craton. In a detailed study, de Kock et al. (2006) separated four superimposed magnetizations and argued that the highest blocking temperature component (HG1) is likely primary, based on a positive fold test, where the folding is interpreted to be contemporaneous with rock formation. However, there is also a negative conglomerate test, in which a magnetization (VAR) with a similar direction overprints the conglomerate clasts, leaving doubt about the age of the HG1 magnetization. The second pole is from Kuetsyvari basalts and andesites in Karelia, thought to have been erupted at ca. 2058 Ma, based on detrital zircons from overlying sedimentary rocks and an intraformational conglomerate (Melezhik et al. 2007). Torsvik and Meert (1995) report a positive conglomerate test for the intraformational conglomerate demonstrating that it has not been remagnetized since deposition. However, the magnetization of the conglomerate clasts is often an order of magnitude more stable than the parent lavas, suggesting that chemical alteration may have taken place. In addition, the poles for the volcanic rocks themselves show significant streaking (Torsvik and Meert 1995, fig. 10) that could indicate unresolved overprinting or apparent polar wander. Further study is required to clarify these issues before the Waterberg and Kuetsyvari poles can be reliably used as key poles.

As previously noted a number of key Paleoproterozoic paleopoles define a rudimentary APWP for the Superior Province. Poles with precise ages between ca. 2.23 and 2.00 Ga are shown in Fig. 11. Paleopoles from the eastern and western Superior Province are slightly discordant and have been interpreted to record a 10°–20° relative rotation of the two portions of the craton across the Kapuskasing structural zone (Halls and Davis 2004; Buchan et al. 2007*a*). Relative rotation occurred after 2.07 Ga, the youngest age for which paleomagnetic data have yet been compared between the eastern and western Superior Province.

Between ca. 2.22 and 2.10 Ga, the Superior Province paleopole moved, in present-day coordinates, from the southern Pacific Ocean, along the western coasts of South America and North America, to the northern Pacific (Fig. 11). Poles from the ca. 2.10–2.00 Ga interval all cluster in the northern Pacific. These include poles for ca. 2106–2101 Ma Marathon reversely magnetized dykes, 2091 ± 2 Ma Cauchon dykes, 2076 $^{+5}_{-4}$Ma Fort Frances dykes, 2069 ± 1 Ma Lac Esprit dykes, and 1998 ± 2 Ma Minto dykes. In sharp contrast, the pole for the 2027–2023 Ma Lac de Gras dykes of the Slave Province lies ~100° to the southeast near Central America, distinct from the 2.10–2.00 Ga cluster of Superior Province poles. In particular, the Lac de Gras pole is far from the Lac Esprit and Minto poles of the eastern Superior Province that are only

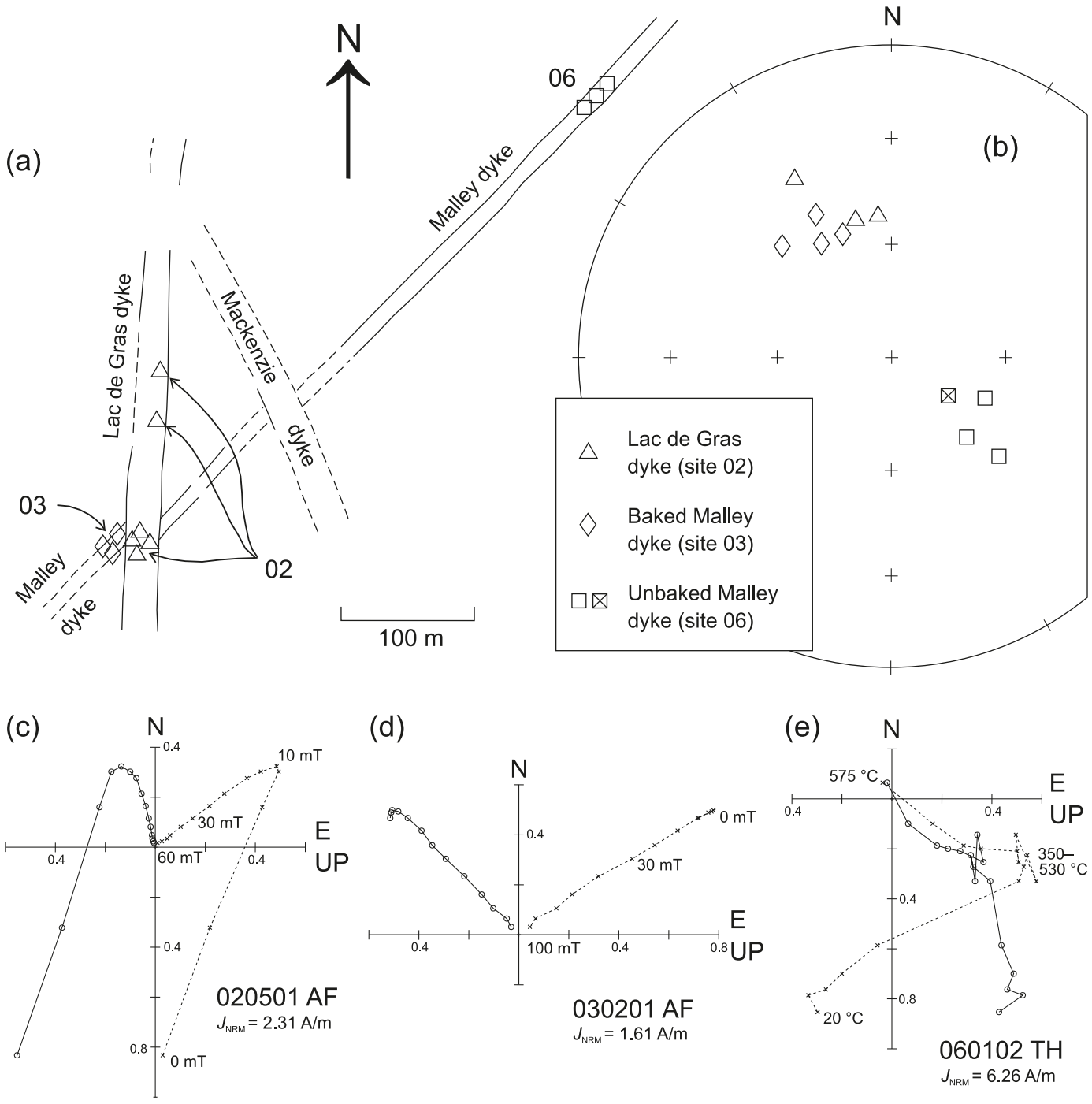
Fig. 9. Baked contact test, where a 1267 Ma Mackenzie dyke crosscuts the 2027 Ma Lac de Gras dyke. (*a*) Sampling locality. Note the offset in the strike of the Lac de Gras dyke caused by intrusion of the Mackenzie dyke. (*b*) Paleomagnetic directions. Mackenzie and baked Lac de Gras directions are down. Unbaked Lac de Gras directions are up. (*c*) Mackenzie sample. (*d*) Baked Lac de Gras sample. (*e*) Unbaked Lac de Gras sample. Symbols in (*c*), (*d*), and (*e*) follow Fig. 5.



~40 million years older and ~25 million years younger, respectively, and far from the Fort Frances pole of the western Superior Province that is ~50 million years older.

In Fig. 12, the latitudinal position (30°–40°) and azimuthal orientation of the Slave Province at 2.03–2.02 Ga, based on the Lac de Gras key paleopole, are compared with the latitude and orientation of the Superior Province in the interval 2.08–2.00 Ga, based on the Fort Frances, Lac Esprit, and

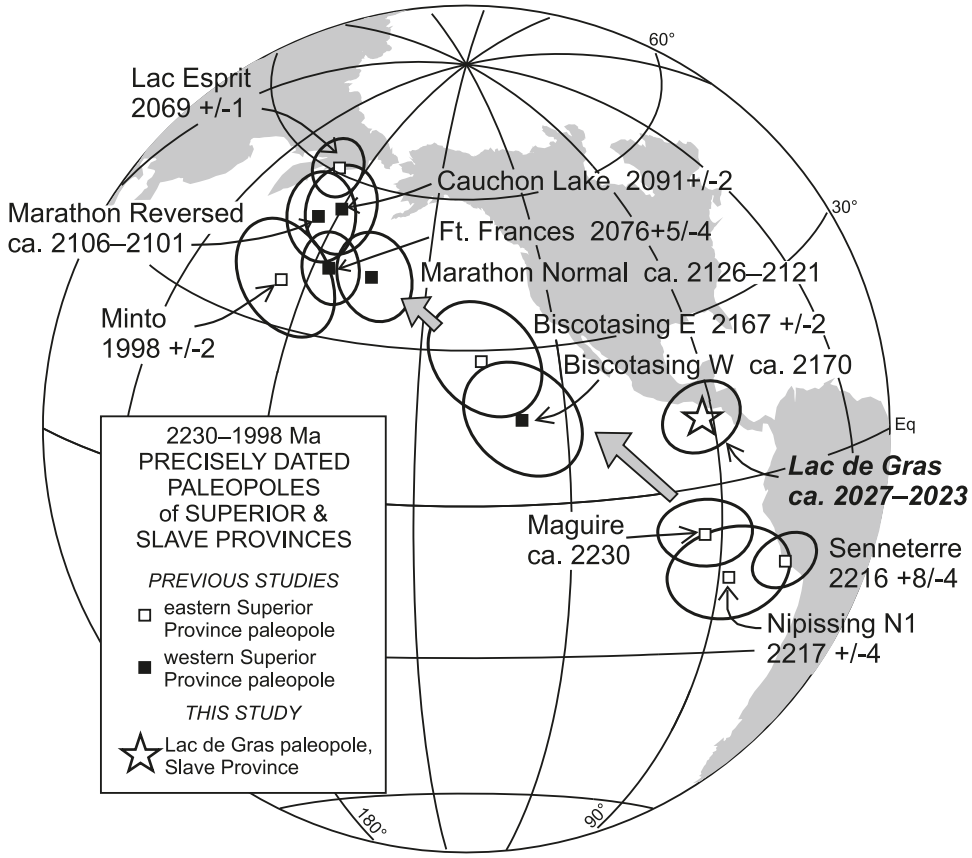
Fig. 10. Baked contact test, where the 2027 Ma Lac de Gras dyke crosscuts a ca. 2230 Ma Malley dyke. (*a*) Sampling locality. Note the offset in the strike of the Malley dyke caused by the intrusion of the Lac de Gras dyke. (*b*) Paleomagnetic directions. Lac de Gras and baked Malley directions are up. Unbaked Malley samples carry up directions with the exception of one down-directed result (square with ×) which has been inverted for comparison with the remaining samples. (*c*) Lac de Gras sample. (*d*) Baked Malley sample. (*e*) Unbaked Malley sample. Symbols in (*c*), (*d*), and (*e*) follow Fig. 5.



Minto poles. Paleolongitude is not constrained. Positions for the Superior Province based on the Fort Frances pole of the western Superior and on the Lac Esprit and Minto poles from the eastern Superior are shown separately, because, as notedearlier, there is evidence that these two portions of the province rotated by 10°–20° relative to one another after the emplacement of these dyke swarms.

Because Slave or Superior APWPs cannot yet be traced continuously back in time from present day to the Paleoproterozoic, the absolute and relative polarities of the Slave and Superior paleopoles used in Fig. 11 are unknown. As a result, two polarity options are possible as illustrated in Figs. 12*a* and 12*b*. In both cases, the Superior Province is located in one (i.e., the same) hemisphere, whereas the

Fig. 11. Precisely dated paleomagnetic poles and their ellipses of 95% confidence for the Slave (star) and Superior (squares) provinces. Open and closed squares indicate poles from the eastern and western parts of the Superior Province, respectively. Ages of poles (in Ma) are based on U–Pb dating of diabase dykes or sills. Grey arrows indicate the time sequence of Superior poles. The Lac de Gras dyke pole and age are from this study; Senneterre dyke pole and age from Buchan et al. (1993, 1996); grand mean pole for Nipissing N1 sills calculated in Buchan et al. (1998); Nipissing sill age from Andrews et al. (1986) and Noble and Lightfoot (1992); Maguire and Minto dyke poles and ages from Buchan et al. (1998); Biscotasing dyke poles and ages from Buchan et al. (1993) and Halls and Davis (2004); Marathon Normal and Reverse dyke poles from Buchan et al. (1996) and ages from Buchan et al. (1996), Hamilton et al. (2002), and Halls et al. (2005); Fort Frances dyke pole and age from Halls (1986) and Buchan et al. (1996), respectively; Lac Esprit dyke pole from Buchan et al. (2007*a*); Cauchon Lake dyke pole and age from Halls and Heaman (2000).



Slave Province is permitted to occupy a position in either hemisphere depending on the polarity adopted. In the first option (Fig. 12*a*), the Slave and Superior provinces lie at similar latitudes of 20°–40°, but have a radically different relative orientation than at present. This permits the two cratons to have formed part of a single continent. However, the Slave and Superior provinces have quite distinct late Archean to Early Proterozoic geological histories (Bleeker 2003). Particularly striking are the differences in ages of pre-1900 Ma Paleoproterozoic mafic magmatic events and giant dyke swarms (Bleeker 2003). This suggests that if the Slave and Superior provinces formed parts of a single continent during this period, they were not near neighbours. Alternatively, they may have drifted separately.

In the second option (Fig. 12*b*), the Slave and Superior provinces are located in different hemispheres and separated by ~40° to 60° of latitude. This option does not permit the two blocks to be adjacent at the time of Lac de Gras dyke emplacement. It suggests that they drifted separately at that time or were located on either side of a large continent or supercontinent. Additional key Paleoproterozoic paleopoles from the two cratons are required to differentiate between these interpretations.

In conclusion, geochronology and paleomagnetism of the Lac De Gras dyke swarm have established the following:

(1) The 2027–2023 Ma Lac de Gras dykes are coeval with the 2026–2023 Ma Booth River igneous complex.

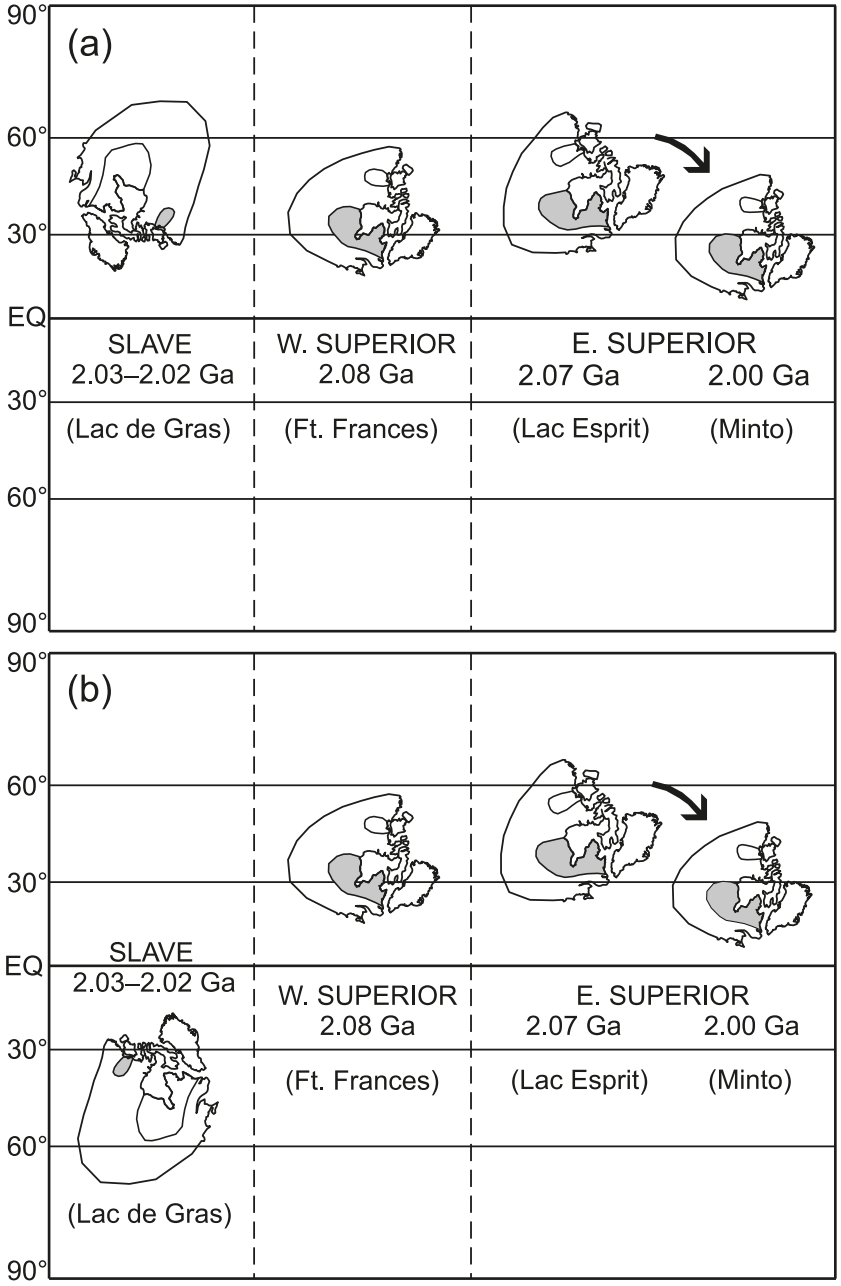
(2) The Slave Province was at a latitude of 30°–40° at the time of Lac de Gras dyke emplacement, based on a key paleopole for the swarm at 11.8°N, 92.1°W (dm = 8.4°, dp = 6.0°).

(3) The Slave and Superior provinces were not in their present relative orientations at that time.

(4) The Slave and Superior provinces were either at similar latitudes (if in the same hemisphere) or separated by 40°–60° of latitude (if in opposite hemispheres).

(5) If in the same hemisphere at similar latitudes, the Slave and Superior provinces may have drifted separately or formed part of a single continent. If they belonged to a single continent, however, geological considerations suggest that they were not adjacent to one another.

Fig. 12. Comparison of paleolatitude and azimuthal orientation of Slave and Superior provinces (shown in grey) at ca. 2.08–2.00 Ga based on key paleopoles. The outline of Laurentia, which is thought not to have existed at that time, is shown to help clarify the orientation of the cratons. Positions of the Superior Province are shown separately for data from the western and eastern Superior, because of evidence that, while remaining adjacent to one another, they rotated 10°–20° relative to one another in the late Paleoproterozoic (Halls and Davis 2004; Buchan et al. 2007*a*). Due to magnetic polarity ambiguity, if the Superior Province is fixed in one hemisphere, the Slave Province may occupy a position in either hemisphere as shown in (*a*) and (*b*). Paleolongitude is unconstrained by the paleomagnetic data.



(6) If in different hemispheres and separated by 40°–60°, the Slave and Superior provinces may have drifted separately or occupied widely separated locations within a large continent or supercontinent.
(7) If Slave and Superior provinces were part of a single continent or supercontinent at 2027–2023 Ma, this continent must have broken up subsequently with the two cratons moving independently to arrive at their present configuration in Laurentia by the late Paleoproterozoic.

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